

Web 3.0 Unveiled - Day 1

Mentored by FEC

What is a Blockchain?

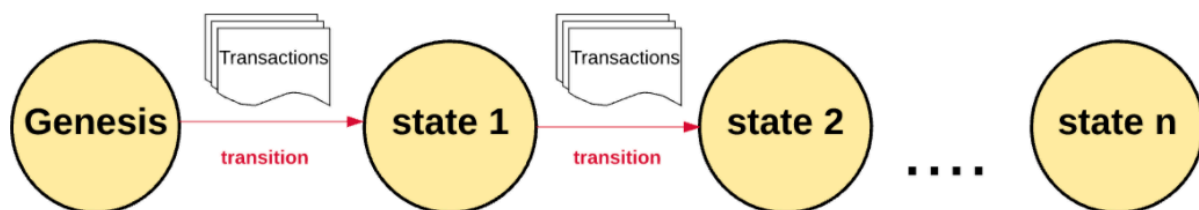
A blockchain is a distributed, digital, immutable ledger that is used to record transactions and store data securely and transparently. It is based on a **decentralized network of nodes** that collectively maintain a single source of truth, allowing participants to interact with the system without intermediaries.

A block is a container data structure that aggregates transactions for inclusion in the public ledger, the blockchain. Each block in the chain contains a set of transactions that have been verified and recorded by the network of nodes through a process known as **consensus**. Once a block is added to the chain, it cannot be altered or deleted, creating an immutable record of all transactions on the network.

Blockchains were popularized by the anonymous person (or group) Satoshi Nakamoto when they released the [Bitcoin Network](#) in 2009. **Bitcoin is a cryptocurrency network** that primarily handles the transfer of BTC assets across the network without a trusted middleman or authority while ensuring the network itself is secure and cannot be hacked. (P.S. **The Bitcoin network is also likely the biggest bug bounty in the world - if you can hack it, you're an instant trillionaire.**).

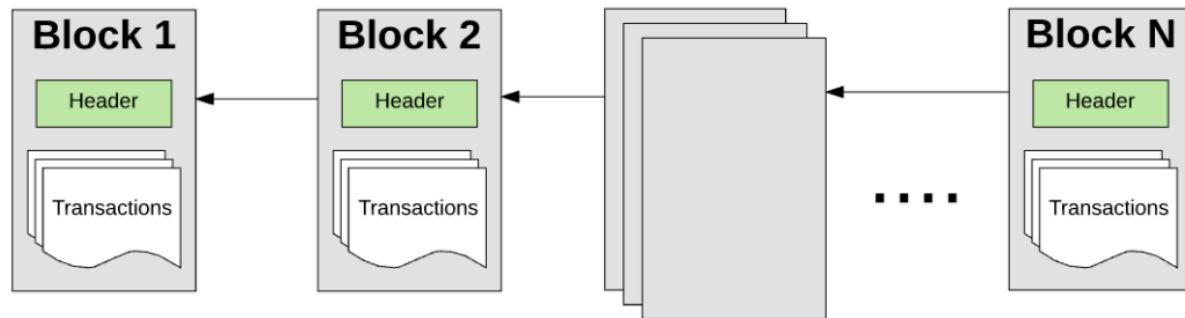
Over time, this design of Bitcoin inspired other, more capable, blockchain networks to emerge, like [Ethereum](#). We will be learning a lot about Ethereum in the coming sessions.

State Management



Blockchains start with a Genesis state when they launch. Bitcoin's genesis state happened in 2009 when the public network launched. Ethereum's Genesis State occurred in 2015 when it launched.

Every transaction on a blockchain modifies the global state that is replicated across all nodes.



Since there are millions of transactions, **transactions get grouped together in blocks**. Hence the name. These blocks are chained together in a cryptographically verifiable way so they are historically traceable. The current state of a network can be recalculated at any time by starting from the genesis block and transitioning the state according to each block's information up until now.

Nodes

A blockchain network is managed autonomously through a [peer-to-peer](#) distributed network of computer nodes. Without going into too much detail, you can think of each node in the network as keeping a copy of the global transaction ledger. Therefore, each node can individually verify and audit transactions happening on the network and ensure there was no illicit behavior.

Another type of node, called a [mining node](#), is responsible for grouping together new transactions being made on a network into a block, verifying them, and proposing the block to be included onto the global ledger by everyone else. **Mining is computationally hard** and very important to do securely, so miners whose blocks get accepted are given a token reward for their hard work.

The use of a blockchain confirms that each unit of value was transferred only once, and the ingenious mechanisms put forth by Satoshi Nakamoto solved the long-standing decentralized [double-spending](#) problem.

Must Watch Video: [📺 How does a blockchain work - Simply Explained](#)